Superlinguistics

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1. Linguistics beyond language

Theoretical linguistics has made great strides towards a proper science of language, building on solid foundations in logic and analytical philosophy, and, more recently, joining forces with empirical sciences. Superlinguistics wants to expand the reach of our sophisticated theoretical and empirical linguistic toolkit by applying it to language-like objects beyond the scope of traditional linguistics, like gestures, comics, or music. Superlinguistics may thus appear to be vulnerable to a charge of 'linguistic imperialism' (Bateman & Wildfeuer, 2014; McDonald, 2013). But the validity of that accusation depends on how you define the original object of linguistic study: language.

Paraphrasing Saussure (1916) we can define a language as a system of signs used for communication. In this definition, 'communication' involves the intentional transmission of information. This excludes forms of what Grice (1957) calls 'natural meaning', like smoke meaning fire. Next, a 'sign' is an object or event that stands for something beyond itself, like the written inscription 'cat' stands for a certain species of animal. This excludes all kinds of human activities and artifacts – a spoon is designed to fulfill a certain function, but it doesn't represent or 'stand for' anything. Finally, we have the notion of a 'system'. This is typically formalized, by linguists, in terms of a grammar (a set of rules generating well-formed signs), coupled with a semantics (a set of rules matching all well-formed signs with what they stand for). This excludes ad hoc representations as in playing with dolls or stage acting.

Linguists traditionally restrict attention to verbal languages in the spoken modality. In the wake of seminal works by Frege (1892), Chomsky (1957), and Montague (1973), many theoretical linguists today are thus primarily concerned with constructing precisely formalized recursive grammars with matching compositional semantic rules that together generate predictions matching native speakers' intuitions. More recently, signed languages have been included as full-fledged languages and objects of theoretical linguistic analysis (Stokoe, 2005), and some subfields also deal with written language (though some others maintain that it is just an artificial representation of spoken language, not a form of natural language worthy of linguistic analysis itself (Bloomfield, 1923; Pinker & Bloom, 1990)).

But there are many different phenomena beyond spoken, written, or signed words that might plausibly still fall under "systems of signs used for communication". A pencil sketch of my cat is a sign that refers to that cat, and as such can be used to *communicate* information about its color or shape. Traffic signs tell you what you should or shouldn't do. Music, hand gestures, facial expressions, and emoji can convey all kinds of subtle information, for instance about their producer's emotional state. To what extent all of these human communication devices are really systematic and/or communicative in the way that verbal languages are is a matter of debate. It is relatively uncontroversial that some genres of instrumental music or martial arts involve following formalizable rule systems, but is there also a semantic system that tells you what sequences of notes or body movements are about? Conversely, photos and other pictures are clearly meaningful, and the semantic relation between a picture and what it depicts can be described rather precisely, but is there an accompanying syntactic system that distinguishes grammatical from ungrammatical pictures? And is smiling governed by any kind of system of syntactic or semantic rules, or just a natural physiological reaction? Superlinguistics studies these types of non-verbal objects in so far as they are indeed so systematic as to be amenable to linguistic analysis, either at the level of form (syntax), or at the level of meaning (semantics), or both (Patel-Grosz et al., 2023; Schlenker, 2018c, 2022).

Brief note on terminology: I use the terms syntax and semantics quite broadly, with syntax encompassing the study of linguistic forms (so including morphology and phonology in addition to (generative) syntax proper), and semantics encompassing the study of linguistic meanings (so including discourse and pragmatics in addition to (formal) semantics proper). I use the corresponding terms supersyntax and supersemantics equally broadly -- and anachronistically, as the use of the 'super'-prefix for this expansion of linguistics is fairly recent, first discussed in print in Schlenker's (2018c) 'What is super semantics?'.[1]

Supersyntax studies the grammar by which signs combine into more complex signs. Examples include studies of the syntactic complexity of bird song (Berwick et al., 2011), syntactic approaches to musical structure and cognition (Lerdahl & Jackendoff, 1983), and more recently to dance movements (Charnavel, 2022) and weight lifting (Esipova, 2022). Supersemantics is the study of how to model the meanings of non-verbal signs, i.e., the relation between forms and what they stand for, and how the meanings of complex signs derive from the meanings of simpler constituents and features of the context of use. Influential examples of supersemantics include studies of the semantics and pragmatics of primate alarm calls (Schlenker et al., 2016), the interaction between gestures and surrounding verbal language (Schlenker, 2018a), and studies of pictorial content and comics (Abusch, 2012). In this chapter I will focus on supersemantics, and more specifically on the formal semantic analysis of multimodal objects.

2. The centrality of multimodality

Signs can convey their meanings in several different ways. Following Peirce (1868) we distinguish first and foremost between symbolic and iconic signs (Dingemanse et al., 2020; Giardino & Greenberg, 2015). Spoken language is highly symbolic, in the sense that most expressions refer to their contents by a somewhat arbitrary, learned system of lexical meaning postulates: 'walking' refers to walking, but only for those who happen to have learned some English. Pictures on the other hand are iconic. The relation between a picture and what it depicts is not based on an arbitrary list of learned conventions, but on a more general notion of resemblance: a drawing of a teapot refers to a teapot because in some sense (that is not trivial or uncontroversial (Goodman, 1976; Greenberg, 2013)) it resembles or shares some geometric or visual properties with an actual (or possible) teapot. In addition to pictures, there are presumably various other iconic sign systems, like certain diagram systems (Giardino & Greenberg, 2015), maps (Greenberg, 2024), onomatopoeia and ideophones (Dingemanse, 2012), demonstration and quotation (Clark & Gerrig, 1990), and music (Schlenker, 2019). Other major sign types beyond symbol and icon have been proposed, like indexicality (my pointing refers to that guy over there, (Kaplan, 1989)), expressivity ('ouch' expresses pain, (Kaplan, 1999)) and exemplification (a paint sample refers to a color, (Goodman, 1976)). I will not attempt to categorize or even list all types of sign systems, also called semiotic modes. I will use the term multimodality to refer to the combination of different semiotic modes in a single sign.[2] Examples include speech with gesture, cartoons with text balloons, and films with sound and music.

Within supersemantics we can study unimodal sign systems like pictures or chord progressions. In this handbook we are interested in multimodal sign systems. These pose additional challenges: not only do we need to describe the meanings of the individual semiotic systems involved, say pictures and text in an advertisement, we also need to characterize how the different forms and meanings combine. We can see in the growing supersemantics literature roughly two types of approaches to modeling multimodal meaning combination: (i) composition at the syntax/semantics interface, and (ii) integration at the level of discourse structure. Let me briefly illustrate these two approaches with some influential formal semantic approaches to gesture, before delving deeper into some case studies in Section 3.

Compositional approaches to gesture semantics (Ebert & Ebert, 2014; Esipova, 2018; Schlenker, 2018a; Tieu et al., 2017) analyze certain co-speech gestures as contributing not-at-issue information that interacts in various ways with the surrounding linguistic context, like embeddings under negation, modals, or quantifiers. Thus, Tieu et al., for instance, argue that embedded gestures semantically entail complex conditionals that incorporate both verbal and gestural content, as illustrated in (1), where <lift> denotes a manual gesture as if the speaker is lifting something heavy, and underlining marks the words during which a co-speech gesture is performed.

(1) John didn't <u>help</u>$_{\text{<lift>}}$ his son.
    ==> if John had helped his son, he would have done so by lifting him

[1] Greenberg (2022) uses the term 'formal semiotics' for roughly what we call superlinguistics here.

[2] Multimodality here does not necessarily require the combination of different sensory modalities or different media, or completely different types of meaning types, as in the film example, where sound and image, and arguably even music, are all iconic, though with different underlying systems of iconicity.

In these types of supersemantic analysis, the gesture's interpretation is assumed to originate inside the logical form of the linguistic sentence (in the VP, embedded under the scope of the negation operator in this case) and then it projects in a certain way, much like other not-at-issue contents, like presuppositions or conventional implicatures (Simons et al., 2011).

Discourse approaches to gesture semantics (De Leon, 2022; Hunter et al., 2018; Lascarides & Stone, 2009) zoom out beyond the grammar and logical form of embeddings and clauses. They model the meaningful interaction between gestural and verbal content at the level of discourse structure. Gestures and verbal utterance alike constitute propositional discourse units that need to be connected to each other via discourse or coherence relations (Hobbs, 1979). Thus, De Leon discusses the inference in (2), where <tattoo> abbreviates the speaker turning their arm to reveal a recent tattoo:

(2) My parents are going to be furious <tattoo>
==> my parents are going to be furious because I got this tattoo

On a discourse-level analysis, the TATTOO gesture conveys that the speaker got this tattoo, and the verbal utterance conveys that their parents will be furious. The interpreter then naturally combines these two propositional discourse units by inferring a causal discourse relation called EXPLANATION (Hobbs, 1979; Kehler, 2002). Crucially, inferring discourse relations between verbal and gestural units follows the same coherence-based reasoning as when both units are verbal (or, as we'll see below, pictorial, in a two-panel comics strip, for instance). In sum, the discourse approach models the crucial chunks of information expressed by various semiotic modes as discourse units and then calls on a pragmatic theory of discourse structure and coherence inferences to combine them, while the compositional approach extends the syntactic/semantic representation formalism of sentence meaning to make room for non-verbal components.

3. Supersemantics in action

I offer four case studies showcasing different aspects of supersemantic approaches to multimodal sign systems. In 3.1 I discuss a compositional semantic approach to the interpretation of facial expressions; in 3.2, I review theoretical approaches to emoji meanings from the recent supersemantics literature; in 3.3, I sketch a possible-worlds semantics for pictures and a discourse-level approach to comics; and in 3.4, I explore how to combine the different iconic semantic contributions of sound, music, and moving images in film in a unified formal semantics framework.

3.1 Facial expressions

The semantic composition of different types of gestural and verbal content has been a core concern of superlinguistics early on (Lascarides & Stone, 2009; Tieu et al., 2017), and if we look outside formal semantics, there is an even longer tradition of interdisciplinary gesture studies (Kendon, 1972). Facial expressions have been considered a type of nonmanual gesturing, but are not as widely studied in semantics – at least, outside of sign language linguistics, where it has been widely recognized that nonmanual markings, including facial expressions, play an important role at all levels of linguistic analysis (Herrmann & Steinbach, 2013; Pfau & Quer, 2010; Wilbur, 2020). Part of the reason for this neglect of co-speech facial expressions is the prima facie plausible thesis, championed by Darwin (1872), that many such apparently meaningful facial expressions are universal, biologically determined reflexes, on a par with sneezing. The intuitive idea that, say, a smile means that you have a positive attitude then would be (at best) a case of natural meaning (Grice, 1957). Since it is then not intentional, it falls outside our broad definition of language as a system of signs for communication (= intentional meaning transmission), and beyond the scope of linguistic methods of analysis.

But consider what happens when I am introduced to a new colleague in the hallway. I smile, say hi, put out my hand, and perhaps engage in a discussion of the dreary Dutch weather. All these actions, including the linguistic utterances describing the weather, happen without much conscious planning or deliberation, but they are not thereby mere physiological reflexes. I clearly intended my actions to communicate a friendly disposition – quite independently in fact of my *actual* mood and interest in the weather. This makes facial expressions, while clearly somewhere on the right side of Kendon's Continuum of gestural semiotics (running from symbolic, via iconic, to natural meaning (McNeill, 2005)), in principle amenable to linguistic analysis. We have to figure out what kind of meaning is expressed by facial expressions, and then investigate how it interacts with other types of verbal and embodied meaning (i.e., compositionally, or at the level of discourse).

The contribution of a co-speech smile may be an expression of happiness (or a related positive emotion or disposition) but standard linguistic tests quickly reveal that it differs from that of an explicit verbal description like “and I’m happy (about that)”. When uttering (3), for instance, my smile would indicate my current positive attitude, while the linguistic paraphrase would be ambiguous, allowing an additional reading where Sue (mistakenly) *thinks* that I am happy.

(3) a. Sue thinks I got in and I’m happy
    b. Sue thi<u>nks I got in</u><smile>

Moreover, when we consider the response in (4), we find that one can (in some contexts) felicitously challenge the explicit assertion that they are happy, but when that positive attitude was instead expressed only by a smile, as in (3b), such a challenge, as in (4), becomes absurd.

(4) That’s not true, you were crying just a minute ago.

Facial expressions also differ from verbal paraphrases in that they are infinitely gradable (from a very subtle raising of a corner of the mouth to a beaming smile involving a variety of facial features), and that the exact type of emotional attitude expressed is highly context sensitive (Ekman et al., 1990). In some of these respects, facial expressions pattern with other non-symbolic signs, like pictures and iconic gestures (“She caught <u>a giant fish</u><arms wide>”), but facial expressions are clearly not iconic or pictorial because they do not refer by virtue of a resemblance: a smiling face does not resemble the abstract concept of happiness or the state of being friendly.

Expressivism is the view that some linguistic constructions can express something without contributing to the derivation of truth-conditional content (Charlow, 2015). What exactly is expressed is a matter of debate, ranging from the emotional state of the speaker (Ayer, 1936; Potts, 2007) to ‘conditions of use’ (Gutzmann, 2015; Kaplan, 1999). On the way to a semantics of face emojis, (Maier, 2023) explores Kaplan’s semantic approach to expressives, which is in turn based on the Wittgensteinian idea that the meaning of (some) expressions is given by the conditions under which they can be used felicitously.

(5) a. An utterance of ‘oops’ is felicitous iff the speaker just observed a minor mishap. (Kaplan, 1999)
    b. A smile is felicitous iff the speaker has a positive disposition towards their addressee. (Maier, 2023)

Use-conditions like in (5) are meant to capture – contra Darwin – that there are ‘correct’ and ‘incorrect’ uses of such expressive words and facial expressions, and it’s the existence of normative rules like these that allow users to intentionally abuse them and deceive unsuspecting interpreters with a seemingly genuine smile. Given such use-conditions, we can define the use-conditional content of a facial expression as the set of possible contexts in which it is felicitously used, just like in traditional formal semantics we define the truth-conditional content of a sentence as the set of possible worlds in which it would be true (Stalnaker, 1970). Gutzmann (2015) gives a detailed extension of compositional semantics to deal with the interaction of use-conditional and truth-conditional contents, which we may adopt for multimodal composition of truth-conditional verbal content and use-conditional face (and face emoji) content.

I end this first case study by mentioning that Ginzburg et al. (2020) take the alternative, discourse-theoretic route, where smiles, sighs, and laughter express propositional contents which are then integrated with verbal propositions in a discourse-level semantic/pragmatic model of conversation (viz., KoS (Ginzburg, 2012)). We’ll encounter the discourse-theoretic route again below, and I’ll provide a more detailed analysis then.

3.2 Emoji

Emojis are an interesting test case for supersemantics. At first sight, the system appears to be relatively simple. Emojis are little, stylized pictures of objects, people, or activities, and their meanings are simply those objects, people, or activities depicted. As for syntax, there’s only a finite number of emoji and they tend to be placed after a written sentence, creating multimodal artifacts in which the two modes, emoji and text, are easily disentangled.

As with gestures and facial expressions we can distinguish between compositional and discourse approaches in the semantic analysis of emoji - text integration. Grosz et al. (2022; 2023) and Pierini (2021) exemplify the compositional approach, at least for the salient subclass of face emojis (😂, 🥰). Grosz et al.’s semantics, for instance, assigns functional lambda expressions to such emojis, fully integrating them into the Montague-style compositional system for deriving truth conditions from sentences (Heim & Kratzer, 1998; Montague, 1973). For object and activity emojis (🍓, 🥊, 🏌), (Grosz et al., 2021; Maier, 2023) take a discourse approach: the sentence expresses a proposition, and the interpreter connects that meaning to the meaning of the emoji via an inferred discourse relation. We'll flesh out the discourse approach here in some more detail below but first we look at the meanings of individual emojis separately.

3.2.1 Emoji semiotics: symbol or icon?

Let's zoom in on the individual emojis and how they express their meanings. Two main options are discussed in the literature: the lexicalist holds that emojis are like new words, related to their meanings via learned lexical conventions, while the pictorialist takes seriously the remark above that emojis are little pictures.

On the lexicalist view, championed by Grosz (2025; 2021), emojis form a symbolic sign system. They move the intuitive iconic character of emoji like 🥦 out of semantics, into the domain of etymology.[3] In Greenberg's (2022) terminology an emoji would then be a so-called 'motivated symbol', just like the verb 'to bark' or the ASL sign for cat (which looks like fingers outlining whiskers on the signer's face) are learned, conventional lexical items, despite their clearly iconic roots. This would fit particularly well with symbol emojis like ❤️ or 💯, but perhaps also for face emojis like 😄 or 😠, which in any case don't resemble or depict what they convey, viz., that the speaker is happy or angry (about what was just mentioned).

On the pictorialist view, championed by Maier (2023), emojis are literally little pictures and thus convey their meanings via resemblance, or more precisely depiction. We'll delve a bit deeper into the semantic analysis of pictorial meaning in Section 3.3 below, but it is quite intuitive that, say, 🏃‍♀️➡️ depicts a woman running and thus conveys that a woman is running. The pictorialist can point to the crucial role of depiction and visual resemblance in the interpretation of skin-tone modified expressives like 👎🏽, non-literal uses of 🍆 (to refer to the penis) and dogwhistle uses of 🥕 (to refer to covid vaccines, (Maier, 2025a)).

Interestingly, in light of our focus on multimodal signs, some emojis could be analyzed as multimodal in that they contain arguably iconic and symbolic elements, like 😍 or 😴. We revisit this and similar types of symbolic enrichment (Maier, 2019) or 'affixation' (Cohn, 2013) in Section 3.3 on comics below.

3.2.2 Emojis in discourse

Next, we consider the inherently multimodal combination of text and emoji. We'll focus here on the discourse approach.

(6) I'm going home 🚲

The interpreter of this multimodal utterance has to first compute the propositional meanings of the two discourse units, the sentence and the emoji. Traditional semantics tells how to derive the sentence's meaning by combining word meanings compositionally. The emoji, construed as a discourse unit, would have to give us a similar type of propositional meaning, like that there's a bicycle or act of cycling somewhere. As discussed above this rather minimal propositional meaning could be derived from a lexical rule or by interpreting the pixel configuration as depicting a bicycle.

Next, we need to infer a discourse relation between the two discourse units to make this into a coherent discourse. In this case, the most coherent discourse involves ELABORATION: the bicycle emoji specifies the mode of transportation for the mentioned trip home. In another context we might infer a different discourse relation:

(7) sorry I have to cancel 🤒

This time the most coherent interpretation involves inferring EXPLANATION: I have to cancel *because* I'm ill.

In addition to the single sentence-final emojis exemplified above, we find longer sequences. Take the following example, which has a salient reading where, roughly, the author was sad and hungry and as a result ate pizza and then went to bed and slept.

(8) really sad and hungry last night 🍕 🛏️ 😴

On the discourse approach outlined above, such a sequence will be interpreted roughly the same as a sequence of sentences in a purely linguistic discourse, viz. via inferred discourse relations:

[3] Or perhaps phonology. One might for instance assume that different emoji sets are different phonological realizations of the underlying semantically symbolic Unicode string (e.g. U?????)

(9) I was really sad and hungry last night [so]$_{\text{RESULT}}$ I had some pizza [and then]$_{\text{NARRATION}}$ I went to bed [and]$_{\text{ELABORATION}}$ fell asleep

Such narrative emoji sequences fit rather well with the multimodal discourse theory's starting point, that emoji composition is driven by our familiar discourse relations and coherence maximization. But it remains to be seen how far we can push this discourse-level analysis in light of the variety of emoji–text integrations that we see today (Tieu et al., 2024).

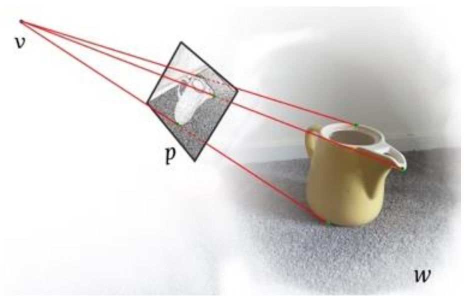
For instance, sequences repeating a single emoji can be used to indicate plurality 🍺 🍺 🍺 or intensity ❤️ ❤️ ❤️, suggesting a kind of 'iconic modulation' (Schlenker, 2018b). It remains to be seen how best to capture the semantics of these kinds of iconic modulation, via discourse relations or compositionally. There are some more intricate ways in which emojis combine with each other and text that so strongly suggest looking inside the grammatical structure within the sentence rather than only the level of inter-propositional discourse coherence: pairs bracketing ✨a piece of text✨ (Kaiser, 2024), beats interspersed 👏 rhythmically 👏 between 👏 words 👏 (McCulloch & Gawne, 2018), and emojis replacing content words in rebus-like fashion (I❤️🍺, Storment, 2024).

3.3 Comics

Comics is a form of visual storytelling that involves putting images in a deliberate sequence. A superlinguistic analysis must give us a precise way to interpret the individual panels, and to describe the way these panels can be combined on the page to tell a story. A salient source of multimodality is the combination of image and text within the panel in many contemporary comics genres, so we will briefly touch on that.

3.3.1 Picture semantics by geometric projection

Following earlier work in semiotics (Eco, 1979; Noth, 1997), linguists and philosophers are starting to converge on the view that pictures, like verbal utterances, express propositions, and can be used to make assertions, and even to lie (Viebahn, 2019).[4] From vision scientists (e.g., in psychology and computer graphics, Hagen, 1986; Szeliski, 2010), we are borrowing precise explanations of how different pictorial representation systems work (Greenberg, 2013). A key concept in the formalization of pictorial content is that of a geometric projection function (*Π*), which we can think of as an algorithm for turning a scene (i.e., a possible world *w* seen from a viewpoint *v*) into a picture (*p*) of that scene: *Π(w,v)=p*. Linear projection, which underlies the familiar idea of linear perspective drawing and photography, works roughly as follows: (i) set up a picture plane perpendicular to the viewing direction of viewpoint *v*; (ii) draw lines from each edge point in *w* to the viewpoint *v*; (iii) put a mark on the picture plane wherever these lines cross it.[5]

(10) 

Different projection functions (axonometric or curvilinear; with color or without; exaggerating or simplifying edges; ignoring or blurring backgrounds, etc.) give rise to different systems, or 'languages', of depiction (Maier, 2026a).

Once we understand how to produce pictures in a certain pictorial language, we can simply take the inverse of that process to get a handle on the semantic interpretation of pictures:

[4] Actually, many philosophers have explicitly argued against the view that pictures express propositional content (Crane, 2009), and some linguists have expressed similar skepticism (Zimmermann, 2016).
[5] Images in (10) and (11) are taken from Maier (2023).

(11) $\left[\!\left[$  $\right]\!\right] = \left\{ w \;\middle|\; \exists v.\Pi(w,v) =$  $\right\}$

This geometric reconstruction of pictorial propositions is a form of iconic semantics (Greenberg, 2022). We can paraphrase (11) informally as saying that the meaning of a picture is the set of possible worlds in which there's a viewpoint from which the world looks like that image – where '*w* looks like *p* from viewpoint *v*' is formalized as 'if we apply the contextually relevant geometric projection function to the scene in *w* that is visible from *v* as input, we get *p* as output'. Our geometric picture semantics is also holistic in the sense that we don't break down the image into smaller constituents, such as objects, shapes, or lines. In this strict sense, then, there is no grammar, syntax, morphology or compositionality in pictorial linguistics (Lande, 2024; Maier, 2026a; Zhang, 2026). Geometric picture semantics differs in this respect from other broadly supersemantic frameworks, like Cohn's (2013) or Wildfeuer's (2019), which do appeal to morphological decomposition of images, for instance in comics panels and emoji. Many of the cases they describe are multimodal symbol/icon hybrids, which are indeed problematic for the inherently holistic approach via geometric projection. A case in point is the hybrid emoji 😍 which we already encountered in Section 3.2. Comics panels provide many instances of such symbol/icon hybrids, from speech balloons and thought bubbles to motion lines and spelled out sound effects.[6]

(12) a. 

b. 

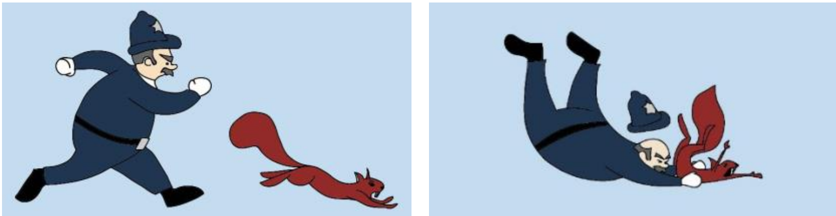
The panel in (12a) pictorially represents a scene of a girl running towards two men on the floor, while the speech balloon linguistically represents the words that she utters. Furthermore, the red stars around the guard's head indicate that he's unconscious. Applying a purely iconic, holistic analysis like that proposed above would (incorrectly, or at least counterintuitively) interpret this panel as depicting a world where elongated white balloons with letters and red stars float over people's heads. Affixation of motion lines as in (12b) is a similar phenomenon, but particularly challenging because it is at least partly iconic, and because it provides comics artists with a way of depicting movement in a static medium, arguably falsifying the thesis that pictures can only depict states (Hacımusaoğlu & Cohn, 2023; Maier, 2026a; Schlöder & Altshuler, 2023).

Maier (2019) proposes a way to extend the holistic projection-based semantics with a decomposition strategy. An image like (12a) is seen as a multimodal complex consisting of (i) a picture, with a balloon-shaped hole, to be interpreted iconically (via geometric projection), and (ii) a speech balloon, to be interpreted symbolically (via a lexical convention of the language of this genre of comics).

3.3.2 Pictures in sequence

Single pictures express propositions, so following the discourse-theoretic approach to supersemantics, we should now be able to create a discourse by putting multiple pictures in sequence. Each picture in such a sequence constitutes a propositional discourse unit, and the interpreter will defeasibly infer various coherence relations to connect them and make the discourse maximally coherent. The example analysis in (13) illustrates what a pictorial extension of the formal semantic framework of SDRT (Segmented Discourse Representation Theory, (Asher & Lascarides, 2003)) might look like, applied to Maier & Bimpikou's (2019) pictorial retelling of a semantics textbook example of a minimal linguistic narrative ("A policeman chased a squirrel. He caught it").[7]

[6] Image in (12a) from *Suske en Wiske 19: De Ringelingschat* by Willy Vandersteen 1951; (12b) from *Garfield 1/10/1991* by Jim Davis, PAWS Inc.
[7] Images in (13) by Sofia Bimpikou, taken from (Maier & Bimpikou, 2019).

(13) a. 

b.

π_1: | $v_1\ x_1\ y_1\ e_1$
v_1: 

policeman(x_1) squirrel(y_1)
chase(e_1)
agent(e_1, x_1) theme(e_1, y_1)

π_2: | $v_2\ x_2\ y_2\ e_2$
v_2: 

$x_2 = x_1\ \ y_2 = y_1$
catch(e_2)
agent(e_2, x_2) theme(e_2, y_2)

NARRATION(π_1, π_2)

The two panels of (13a) show what the world looks like, from two spatiotemporal viewpoints. In the SDRT logical form (13b) we find the interpretation of the individual panels as boxes labeled π_1 and π_2; π_1 semantically contributes the following: (i) there exists a viewpoint v_1, two entities x_1 and y_1, and an event e_1; (ii) from v_1 the world projects onto the picture; (iii) from v_1 the entity x_1 looks like that dark-blue tagged region etc.. By pragmatic enrichment we add: x_1 is a policeman and y_1 a squirrel (and similar for π_2). The final component of the SDRT logical form is the inferred discourse relation, NARRATION, that holds between these two discourse units.

When we look at wordless picture books or comics through the lens of discourse structure theory, we will see a lot of NARRATION relations: first the event depicted in panel 1 happens, *and then* the event in panel 2, and so on (see (McCloud, 1993) pioneering discussion of panel transitions). Related work in film semantics has uncovered additional constraints on narrative transitions like these (Cumming et al., 2017; Huff & Schwan, 2012). For instance, within a scene, the viewpoint tends to stay on one side of the most salient action, i.e., the camera position may change from panel to panel, but it never crosses the line of action.[8] Such a filmic convention helps viewers in mentally reconstructing a coherent space, presented by a sequence of shots from different angles. Semantically, we could formalize this and other conventions as viewpoint constraints that are baked into the semantics of certain coherence relations, like NARRATION (Maier, 2025b).

As in written or oral narratives, we do not have to look very far to find other types of coherence relations. In (14) we see two panels of children praying, presumably in different parts of the world and without any suggestion that one prayer temporally precedes the other – a configuration we might analyze in terms of the coherence relation PARALLEL.[9]

[8] Walking, driving, punching, but also looking at something or talking to someone count as actions with an inherent direction.

[9] Comic in (14): *Spelling* by Christophe Frochaux, Perry Bible Fellowship. https://pbfcomics.com/comics/spelling/.

(14)

3.3.3 Perspective

The representation of different perspectives is arguably as essential to good storytelling as temporal sequencing of events via NARRATION. In visual as in verbal narrative there are various conventions for conveying subjective perspectives of different characters in the story. In verbal narrative we have different grammatical constructions, like quotation and (free) indirect discourse. In comics we have a variety of more or less morphologically marked constructions as well (Klomberg et al., 2023). First, we have thought bubbles that represent a character's thoughts, dreams, or imagination verbally or pictorially. As discussed above in 3.3.1 we can treat the bubble shape itself as the visual--morphological realization of a kind of quotation operator, akin to written quotation marks (Maier, 2019).[10]

(15) a. 

b. 

Next, we have point of view shots or free perception sequences (Abusch & Rooth, 2023), where, typically, one panel depicts a person looking, and the next depicts what they see, from their perspective, i.e. as if through their eyes.[11]

(16) 

[10] Image in (15a) from *Garfield 1/1/1981*; (15b) from *Garfield 2/27/1981* by Jim Davis, PAWS Inc.

[11] Image in (16) from *PhD Comics* by Jorge Cham, 2018. https://phdcomics.com/comics/archive.php?comicid=2030.

A third common convention involves a 'blending' of perspectives: what is depicted is the world as subjectively experienced by one of the characters, but the viewpoint is that of an objective narrator, often allowing us to see parts of the scene that the experiencing character would not be able to see (Maier & Bimpikou, 2019), as illustrated in (17), which depicts the nightmarish visual hallucinations that Robin is subjectively experiencing, but simultaneously depicts Robin's own back as he is tied to a chair.[12]

(17) 

In analyzing these types of perspective shift, Abusch & Rooth (2017) go the compositional route. A hidden operator in the logical form of the picture is responsible for the shift. Maier (2026b) takes the discourse route, appealing to the inference of a discourse relation of ATTRIBUTION to connect a representation of someone experiencing a mental state in one unit, with a representation of the content of that mental state in another.

3.4 Film, sound, and music

3.4.1 Moving images

Film, or video, is a ubiquitous medium for vivid storytelling. Most instances today are multimodal, from 30 second TikToks to Hollywood movies. We'll briefly review some supersemantics work treating film as a form of visual narrative, and add some reflections on the role of sound and music.

Film may be defined as a sequence of shots edited together to present a coherent sequence of events. A shot in turn is defined as a moving image that corresponds to a continuous camera recording (Cumming et al., 2017). Technically, a moving image is often created out of a sequence of still images, which in turn consist of pixels etc., but since human viewers do not perceive it as such, we can safely ignore these further decompositions: from a semantic point of view the shot is the basic unit of information in film, like the panel is for comics, and the sentence or clause for verbal languages.

A central question of film cognition is how viewers manage to extract a coherent story from the sequence of shots they are presented with. Some film theorists have posited that there is a conventional 'film language' that viewers and filmmakers have converged on (Metz, 1990). In light of our discussion of the different approaches to superlinguistics we might see this as an early instance of the compositional approach, modeling non-verbal sign combination at the syntax/semantics interface. Philosophers have pushed back against this type of approach, treating film cognition instead as continuous with natural perception, i.e., we make sense of film in the same way we make sense of what we see in the world around us (Currie, 1995). More recently, semioticians (Bateman, 2007; Wildfeuer, 2014) and semanticists (Cumming et al., 2017) have advocated an intermediate, discourse-level approach, similar to the proposed discourse analysis of comics in Section 3.3 above.

On a discourse approach, the first task is to develop a semantics for the basic discourse unit – the shot. We start with a straightforward extension of the geometric projection-based semantics for pictures: a viewpoint is now no longer simply a vector (the viewing direction) located in space and time, but rather a vector continuously moving through space over a given time interval. We get logical forms that contain

[12] Image in (17) from *Batman and Robin Eternal #2*, 2016 by various artists, DC Comics.

variables representing moving viewpoint vectors and also actual shots – just like the pictorial SDRT logical forms we saw in (13b) for comics contain actual pictures.

3.4.2 Audio projection

A novel set of challenges posed by film shots as opposed to panels comes from the audio. Film shots today often come with various types of sounds. Diegetic sounds (i.e. sounds that are part of the narrative world), including dialogue, can be treated as closely resembling the sounds that would be heard in the fictional world by a hypothetical observer located at the (moving) viewpoint. Formally we just introduce an audio equivalent of projection, taking a world and moving viewpoint as input and a sound recording in some digital or analogue medium as output. The mathematical details of how sound gets 'projected' from a microphone onto a magnetic tape or an mp3 file need not concern us here.

(18) a. $\Pi_{video}(w,v)$ = recording of visual scene (e.g., as a sequence of frames) picked up by a camera at v in w.

b. $\Pi_{audio}(w,v)$ = recording of sound waves (e.g., as a sequence of sound pressure levels) picked up by a microphone at v in w.

Let's model a shot S as a pair consisting of a moving image S_{IM} and an audio recording S_{AU}. The possible-worlds semantic interpretation of S is given by inverting the projections in (18), giving us those possible worlds that 'look and sound like that' (from some camera-with-microphone viewpoint v):

(19) $[\![S]\!] = [\![< S_{IM}, S_{AU} >]\!] = \{ w \mid \exists v \,.\, \Pi_{video}(w,v)=S_{IM} \;\&\; \Pi_{audio}(w,v)=S_{AU} \}$

Extra-diegetic sounds, including music and voice-over, are more like the emotional affixes and text balloons in comics in that their interpretation relies on very different, non-projective sign systems, and more complicated integration -- e.g., the music or voice-over might represent aspects of a character's subjective, inner mental state, over a sequence of shots that show that character (or other characters) from an apparently objective narrator perspective. I'm unaware of in-depth formal semantic analyses of this multimodal combination, but in the final subsection below I will sketch out the basics of a formal semantics for music that at least allows us to interpret music as having truth conditions and expressing possible worlds propositions, bringing it in line with our semantics of video and audio and thus in principle allowing us to start modeling this complex multimodal composition.

3.4.3 Music semantics

Music has been a fruitful area of study in superlinguistics since Lerdahl & Jackendoff's (1983) seminal work. Their application of generative linguistics to music theory is a prime example of what we might now call supersyntax. It focuses on how we parse music, i.e., how listeners create hierarchically structured representations, similar to linguistic tree structures, from a piece of music, using a set of rules about melody, harmony and rhythm. Like Chomskyan syntactic theory, it doesn't really provide a semantics, i.e., it doesn't capture what, if anything, a piece of music is about, what a composer communicates to her listeners about some extra-musical reality.

Schlenker (2017, 2019) develops a supersemantics approach to musical meaning. A piece of music conveys a story about something (a 'virtual source') moving and changing over time, e.g., the source may move slowly or excitedly, come closer or move away, and go from a stable resting position to a more unstable state or the other way around. This type of musical meaning is conveyed iconically: the 'voices' in a piece move in a complicated multidimensional musical space and aspects of these movements resemble or correspond to (in some sense, to be made precise) aspects of the movements and changes of the corresponding virtual source moving around in the extra-musical world.

Concretely, Schlenker's music semantics spells out a number of constraints on the resemblance relation between musical movements and source movements. For instance, a voice that increases in loudness means that the source is coming closer, and a voice with a lower pitch denotes a larger source than a voice with a higher pitch. Building on well-established insights from Western, classical music theory, Schlenker includes also more specifically musical constraints in his music semantics. For instance, a voice that moves from the dominant (the fifth note or chord of the relevant scale in use) to the tonic (the first, primary note or chord of the scale), corresponds to a source that changes from tension to rest.

The core of Schlenker's semantics is a definition of musical truth, which gives rise to a familiar possible worlds proposition. Let's consider a piece of music M with a single voice, consisting of a temporally ordered sequence of musical events $M_1,\ldots,M_n$. The music then denotes the set of worlds where we can find some entity s, a virtual source, that undergoes a sequence of events $e_1,\ldots,e_n$ that mirror the musical movements:

(19) $[\![M]\!] = [\![\langle M_1,\ldots,M_n\rangle]\!] = \{ w \mid \exists s\, \exists e_1,\ldots,e_n \,.\, e_1,\ldots,e_n$ is a temporally ordered sequence of events in w & s participates in all of $e_1,\ldots,e_n$ in w & $\forall i<n$. (if M_i is louder than M_{i+1}, then s is closer or more energetic in e_i than in e_{i+1} & if M_i is harmonically more stable than M_{i+1} then s is in a more stable position in e_i than in e_{i+1}) $\}$

The next step towards a multimodal supersemantic analysis of film would be to align the projective audiovisual film shot semantics in (18) with the music semantics in (19). This would entail, among other things, introducing musical and audiovisual discourse referents, pragmatically equating referents representing virtual sources with audiovisual discourse referents, and making room for modality-specific intensional operators (for instance, to interpret the music as describing the emotional movements occurring inside the visible protagonist's mental state). I leave this for future research.

4. Conclusion

Superlinguistics uses the toolkit of traditional theoretical linguistics, which in turn builds on logic and philosophy, and applies it to sign systems beyond the traditional objects of linguistic analysis, i.e., natural languages like ASL, spoken Dutch, or written English.

In this overview I have discussed a few concrete examples of superlinguistic analysis of such sign systems. In my selection I have restricted attention to supersemantic analyses of multimodal signs. I have not tried to be comprehensive in my review, but I have also not restricted attention to approaches that explicitly self-identify as falling under the header of superlinguistics. A recurring theme in these case studies has been the split between compositional approaches and discourse approaches to multimodal supersemantics. The former model the integration of different sign systems by extending the lexicon and/or modifying semantic composition rules; the latter by generalizing how we build a coherent discourse structure out of propositional units.

Let's return now to the charge of 'linguistic imperialism' from the start, i.e., the charge of treating everything as a nail, just because you have a good hammer. I end this overview with three responses. First, if we look at the origins of modern linguistics we find there already a quest for a general theory of signs (Saussure, 1916). Superlinguistics then is just a return to the original program, where the objects of analysis should include any system of signs used to communicate. Second, some verbal and non-verbal sign systems are so interconnected that studying just one in isolation, say, just signed or spoken sentences without the accompanying iconic and deictic gestures, cannot lead to a satisfying analysis of linguistic meaning or communication (Goldin-Meadow & Brentari, 2017). Third, the aim of superlinguistics is not just to expand the subject matter of linguistics, but also to take any insights gained from that expansion and bring them back to the original verbal linguistic domain. A case in point is the semantic analysis of comics informing the analysis of verbal aspect, perception reports, and attitude ascription (Abusch, 2014; Maier, 2026b). More generally, the eventual supersemantic analyses of non-verbal sign systems tend to highlight not only the similarities with verbal languages (e.g., propositional segments in any modality are connected with discourse relations to maximize coherence), but also bring out the essential differences (e.g., holistic projection for pictures vs compositional semantics for English), thereby teaching us also what is unique about language and giving insight into the relation between language and communication.

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